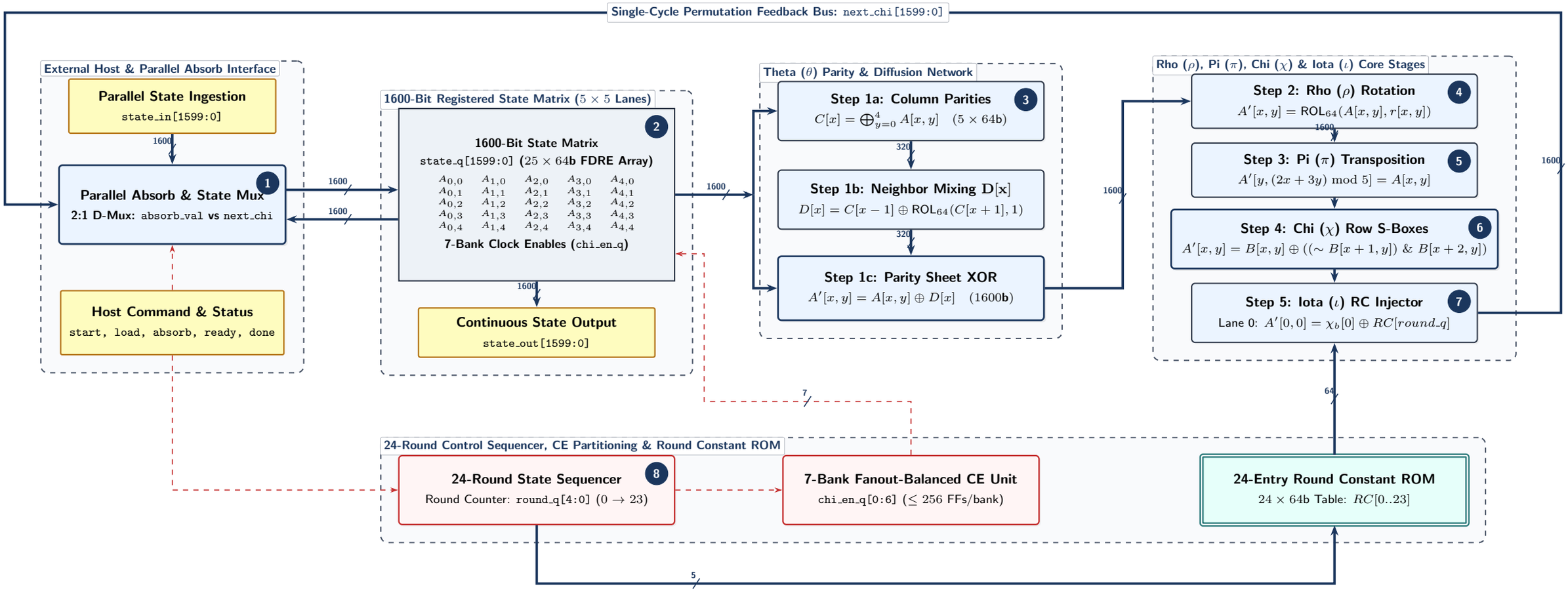
ML-KEM HPKA: 24-Cycle Keccak-$f[1600]$ Permutation Engine

Subsystem 08 Microarchitecture — RTL Module: `keccak_f1600.sv` (`FAST_24=1`, Single-Cycle Round Pipeline)



ID	Component / Signal	Width / Format	Description
1	Parallel Absorb & State Mux	1600-bit Data / Ctrl	2:1 synchronous injection mux; loads `state_in` directly (`load=1`) or absorbs in-place into `state_q` in 1 cycle.
2	Registered State Matrix (`state_q`)	1600-bit (25×64b)	5×5 array of 64-bit lanes (1,600 FDRE flip-flops); continuously drives `state_out[1599:0]` at 450 MHz.
3	Theta (θ) Parity Diffusion Network	5×64b $\rightarrow$ 1600b	Computes column parities $C[x]$, neighbor mixing $D[x]$, and XORs into all 25 lanes in 2 LUT levels.
4	Rho (ρ) Bitwise Rotator Array	24×64b Intra-Lane	Bitwise cyclic left rotations on 24 lanes by fixed offsets $r[x, y]$; implemented as zero-cost wire routing.
5	Pi (π) Coordinate Transposition	25×64b Inter-Lane	Lane coordinate permutation mapping $y \leftarrow 2x + 3y \bmod 5$; fused with ρ as pure interconnect routing.
6	Chi (χ) Row S-Box Layer	1600-bit Non-Linear	5-bit quadratic non-linear S-box array operating across each row; maps into a single LUT level.
7	Iota (ι) Injector & RC ROM	64-bit Asymmetry Bus	XORs 64-bit LFSR round constant $RC[round_q]$ into origin lane $A[0, 0]$ to break permutation symmetry.
8	24-Round State Sequencer & CE Unit	Control / 5-bit Counter	Governs single-cycle round sequencing ($0 \rightarrow 23$, 24 cycles total) and drives 7-bank clock enables (≤ 256 FFs/bank).